

Paper II — The Photon as a Bound Pair of Spin- $\frac{1}{2}$ Lattice Quanta: A Non-Birefringent, Luminal, Transverse Composite

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Notebook source material: M. Ludwig (2007), *Physics Notes*, pp. 5–6 (“spinor electrodynamics”)

Series: “A Universe in a Bottle” — Paper II of XI. Builds directly on Paper I (BCC QCA substrate; speed of light as rotation rate).

Revision 2 (2026-06-08): re-issued for the eleven-paper series; cell-size statements updated to the canonical F107 value $a = 6.5978 \ell_P$ adopted in Paper VII.

Abstract

We construct the electromagnetic photon not as a fundamental spin-1 gauge boson but as a bound pair of two spin- $\frac{1}{2}$ Weyl quanta of the underlying body-centred-cubic (BCC) quantum cellular automaton — the lattice realisation of de Broglie’s neutrino theory of light. The pair carries total momentum $\mathbf{k}$ shared as $\mathbf{k}/2$ between two constituents on opposite chiral branches, so its real $(\mathbf{E}, \mathbf{B})$ field rotates at the **helicity-symmetric** rate $\Omega_{\text{pair}}(\mathbf{k}) = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) \equiv \Omega_{\text{even}}(\mathbf{k})$. We show this object is (i) massless and luminal, with group velocity $\rightarrow 1/\sqrt{3} = c_{\text{lat}}$; (ii) transverse and spin-1, with two physical polarisations; and (iii) **exactly non-birefringent**, because both helicities ride the single rate Ω_{pair} — there is no single-branch photon to split against. This last property is decisive: an earlier composite construction (the σ -bilinear photon) assigned each helicity to its own branch and was therefore birefringent at first order in k , a prediction excluded by gamma-ray-burst (GRB) and active-galactic-nucleus (AGN) polarimetry by many orders of magnitude at the gravitationally-fixed cell size. The paired photon removes that prediction entirely and is independently forced from three directions — observation, $U(1)$ minimal coupling, and Ludwig’s notebook construction — that all land on the same dispersion. All claims are verified to machine precision (5/5 checks; residuals 0 to $\sim 10^{-15}$).

1. Introduction

Paper I established that on the BCC QCA a correlated pair of lattice spinors furnishes a real field pair $(\mathbf{E}, \mathbf{B})$ whose free evolution is an exact rotation, and that the speed of light is the rotation rate of that pair per unit wavenumber. The present paper answers the next question: *what, precisely, is the photon?*

In the Standard Model the photon is a fundamental spin-1 gauge field A_μ introduced by hand to gauge $U(1)$. On the lattice no such field need be posited. Following a proposal recorded in Ludwig’s 2007 notebook (pp. 5–6) — and ultimately de Broglie’s 1930s neutrino theory of light — the photon is a **bound pair of two spin- $\frac{1}{2}$ “fermion-photons” $\gamma_{1/2}$ that behave together as a single boson and only occur as a pair; they do not occur separately**. This is structurally the same idea as a Cooper pair in superconductivity: two spin- $\frac{1}{2}$ objects bound into a single bosonic excitation.

The composite photon arises *natively* from the lattice's own Weyl quanta and requires no additional field construct, which is its chief economy over the Standard-Model photon. The task of this paper is to make the construction precise, derive its propagation law, and confront its observational consequences.

2. The two-branch structure of the BCC walk

From Paper I (Eq. 2.1) the free Weyl walk has two dispersion branches,

$$\omega^\pm(\mathbf{k}) = \arccos(c_x c_y c_z \pm s_x s_y s_z), \quad c_i = \cos \frac{k_i}{\sqrt{3}}, \quad s_i = \sin \frac{k_i}{\sqrt{3}}, \quad (2.1)$$

related by parity $\omega^+(-\mathbf{k}) = \omega^-(\mathbf{k})$. The two branches are the two chiralities/helicities of the walk. A real electromagnetic field is built from the Riemann–Silberstein (RS) combinations

$$\mathbf{F}_\pm(\mathbf{k}) = \mathbf{E}(\mathbf{k}) \pm i\mathbf{B}(\mathbf{k}), \quad (2.2)$$

which are exactly the eigenvectors of the one-tick rotation matrix $R(\Omega)$ of Paper I: $R(\Omega)(1, \mp i)^\top = e^{\mp i\Omega}(1, \mp i)^\top$. Reality of $(\mathbf{E}, \mathbf{B})$ forces $\mathbf{F}_+(-\mathbf{k}) = \mathbf{F}_-(\mathbf{k})$.

2.1 The fork: two ways to assign helicities to branches

There are two consistent ways to propagate the RS eigenstates while preserving the reality (Hermitian-symmetry) constraint:

- **Chiral assignment (the σ -bilinear photon, F29/F37/F39):** let each helicity ride its own branch, $\mathbf{F}_+ \rightarrow \Omega^+ = 2\omega^+(\mathbf{k}/2)$ and $\mathbf{F}_- \rightarrow \Omega^- = 2\omega^-(\mathbf{k}/2)$. Hermitian symmetry is preserved because $\Omega^+(-\mathbf{k}) = \Omega^-(\mathbf{k})$.
- **Even assignment (the paired photon, this paper):** let *both* helicities ride the single symmetric rate $\Omega_{\text{pair}} = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) = \Omega_{\text{even}}$.

These two laws are physically inequivalent, and §4 shows the first is observationally dead while the second survives. The chiral law is retained only for the massive/confined sectors (Papers V–VI), which carry no astrophysical vacuum-birefringence bound. The systematic statement of which sector rides which law is the pairing classification theorem of Paper IV (Finding F91): the propagator law is *forced* by the chirality structure of each coupling, not chosen.

3. The paired photon

3.1 Construction

The photon is a bound $(+, -)$ pair: total momentum $\mathbf{k}$ shared by two Weyl constituents at $\mathbf{k}/2$ each, one on the $+$ branch and one on the $-$ branch. Because the pair is bound and symmetric, the phase it accumulates per tick is the **sum** of the constituent phases:

$$\boxed{\Omega_{\text{pair}}(\mathbf{k}) = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) \equiv \Omega_{\text{even}}(\mathbf{k}).} \quad (3.1)$$

This is exactly the even-law rotation rate of the lattice’s $(\mathbf{E}, \mathbf{B})$ propagator from Paper I — no new propagator is invented; the pairing *names* the even law as the photon law and supplies its physical interpretation. The free field then evolves by the exact rotation (Paper I, Eq. 3.1) with $\Omega = \Omega_{\text{pair}}$. The propagator is implemented as `ca_photon_pair.py` (propagator = even law `ca_wmu._f26_rotation_step`).

3.2 Why non-birefringent — the mechanism

In the chiral construction the two photon helicities $\mathbf{F}^\pm$ were assigned to the two branches independently, so a generic linear polarisation (a superposition of $\mathbf{F}^+$ and $\mathbf{F}^-$) acquired a relative phase $\Delta\Omega N$ after N ticks and its plane of polarisation rotated — vacuum birefringence. In the **paired** photon there is no single-branch photon: it “only occurs as a pair,” always containing both branches symmetrically, so both helicities of the resulting field ride the *one* rate Ω_{pair} . The split has nothing to split against. Formally, the birefringence

$$\Delta\Omega(\mathbf{k}) = \Omega^+(\mathbf{k}) - \Omega^-(\mathbf{k}) \quad (3.2)$$

which is non-zero for the chiral law (see §4), never enters the paired-photon evolution because the pair is built symmetrically in $\pm$ from the start.

3.3 Massless, luminal, transverse, spin-1

Direct evaluation of (3.1) gives:

- **Massless and luminal:** $\Omega_{\text{pair}}(0) = 0$ and $\Omega_{\text{pair}}/|\mathbf{k}| \rightarrow 0.577350 = 1/\sqrt{3}$, with group velocity $d\Omega_{\text{pair}}/d|\mathbf{k}| \rightarrow 1/\sqrt{3} = c_{\text{lat}}$ (verified to 7.7×10^{-6} by finite difference; the slope along the body diagonal evaluates to 0.5773503).
- **Transverse, spin-1:** the field has two transverse polarisations, $|\mathbf{E} \cdot \hat{\mathbf{k}}|/|\mathbf{E}| = 1.8 \times 10^{-17}$; it is real (10^{-15}) and norm-conserving (6×10^{-15}).
- **“Only as a pair”:** the pair phase equals the constituent sum $\omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2)$ exactly (residual 0); an unpaired single branch would carry $\Omega^\pm$, which differs (median split 5.3×10^{-3}).

The two transverse polarisations and the masslessness together give the photon its spin-1 content without any spin-1 field being fundamental: the spin-1 object is the symmetric combination of two spin- $\frac{1}{2}$ constituents ($\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$, with the antisymmetric singlet being the pseudoscalar partner — the pion-channel sibling of Paper X).

4. Confrontation with polarimetry: why the chiral photon is excluded

Along the BCC body diagonal (1, 1, 1) the chiral-law birefringence is, to leading order (expansion of (2.1)),

$$\Delta\Omega \approx -\frac{\sqrt{3}}{27} k^2, \quad \frac{\Delta v_\phi}{c_{\text{lat}}} \approx -\frac{k}{18}, \quad (4.1)$$

i.e. **linear vacuum birefringence** in photon energy ($k \propto E_\gamma/E_{\text{Planck}}$); it vanishes on the cube axes and is maximal along the body diagonals. Linear birefringence rotates the polarisation angle

of light from a distant source by $\Delta\theta \sim \Delta k \cdot d$, and is tightly constrained by GRB and AGN polarimetry (INTEGRAL, *Fermi*/GBM). At the cell size that the gravity sector independently requires (Paper VII; $a = 6.5978 \ell_P = 1.07 \times 10^{-34}$ m), the chiral law overshoots the polarimetry bound by many orders of magnitude — it is excluded precisely at the gravitationally-fixed cell size (Findings F65/F66).

The paired photon (3.1) removes this prediction entirely: there is no helicity split, so a linearly polarised body-diagonal mode shows the symmetric phase $S = \phi_+ + \phi_- = -1.8 \times 10^{-15}$ (exactly non-birefringent) where the chiral law gives $S = +0.0792 = -\Delta\Omega N$. This is the decisive empirical reason the paired photon supersedes the σ -bilinear as *the* electromagnetic photon (Finding F67).

5. Three independent routes to the same object

The paired photon is not an *ad hoc* repair. It is forced from three independent directions that now agree:

1. **Observation (§4):** the photon must be non-birefringent $\Rightarrow$ it must ride the helicity-symmetric rate Ω_{even} .
2. **$U(1)$ minimal coupling (Paper IV; Finding F68):** the charge-coupling photon is the $U(1)$ *identity channel* $e^{i\theta}\mathbf{I}$, which acts identically on both helicities — helicity-blind, hence Ω_{even} . The general statement (F91) is that the unique identity-channel direction of any neutral coupling $aT_3 + bY/2$ is the one proportional to electric charge Q ; that direction *is* the photon.
3. **Ludwig’s notebook (pp. 5–6):** the photon is a bound pair of two $\gamma_{1/2}$, symmetric in $(+, -)$, hence rate $\omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) = \Omega_{\text{even}}$.

All three land on the same dispersion. The notebook’s open question “Is there a scalar photon?” (p. 6) is thereby answered: the scalar/identity photon **is** the paired photon, and it is the physical one.

6. Consistency with the gravity sector

The non-birefringence of the paired photon does more than settle a polarimetry tension: it **restores the cell size the gravity sector requires** (Paper VII). The structural (Sakharov-induced) value of Newton’s constant pins the cell at the parameter-free value $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P$ (Finding F79/F107). The chiral photon was falsified *at that very cell size*; the paired photon survives all astrophysical vacuum-birefringence bounds there, with the Lorentz-violation ($n = 2$) cutoff at $E_{\text{QG},2} = \sqrt{54} \hbar c/a \approx 1.36 \times 10^{19}$ GeV — above the current GRB/LHAASO time-of-flight bounds. The model’s dimensionless predictions (Weinberg angle, mass ratios, $c_{\text{lat}} = 1/\sqrt{3}$, PPN $\beta = \gamma = 1$, Mercury precession, factor-2 light bending) are cell-size-independent and unchanged; what the paired photon buys is that the gravity-fixed a is no longer ruled out by the photon sector. This co-certification ties Papers II, VII and VIII together.

7. Verification summary

#	Check	Result	Status
PP1	$\Omega_{\text{pair}} = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2) =$ even-law rate (2000 modes)	residual 0	exact
PP2	Non-birefringent: linearly polarised body-diagonal mode, 6 ticks	paired $S = -1.8 \times 10^{-15}$ vs chiral $S = +0.0792$	machine
PP3	Massless & luminal: $\Omega_{\text{pair}}/k \rightarrow 0.57735$, $v_g \rightarrow 1/\sqrt{3}$	err 7.7×10^{-6}	quantitative
PP4	“Only as a pair” (constituent sum vs single-branch split)	residual 0	machine
PP5	Spin-1, two transverse polarisations, real, norm-conserving	$\leq 6 \times 10^{-15}$	machine

Underlying findings: F69 (paired-spinor photon), F67/F68 (even-law vs bilinear; minimal coupling forces the even photon), F65/F66 (helicity–chirality map; all-sky birefringence bound), F37 (RS/BCC chirality), F39 (the retired two-helicity bilinear), F91 (pairing classification theorem).

8. Discussion

What is gained. The photon is now an emergent composite of the lattice’s own quanta, with no new fundamental field. Its masslessness, luminality, transversality and — crucially — its exact non-birefringence are all consequences of the symmetric pairing, not inputs. The construction unifies an observational constraint, a gauge-theoretic requirement, and a 2007 notebook conjecture.

What is retained for other sectors. The chiral σ -bilinear is *not* discarded wholesale: it remains the correct *field construction* for the massive and non-Abelian sectors — W , Z , and gluons (Papers V, VI) — which are massive or confined and carry no astrophysical vacuum-birefringence bound. Indeed the pairing classification theorem (F91) shows the propagator *law* (even vs chiral) is forced sector by sector: γ even (forced), $W^\pm$ chiral (forced; left-projector coupling), Z even for its vector part with a mass-suppressed axial split, gluon even (forced; colour coupling is branch-blind).

Open items. (i) A genuine two-constituent bound-state simulation of the pair — the “negative binding energy” of the notebook and the massless-pair condition — beyond the present dispersion-level treatment (partly addressed by the bound-state machinery of Findings F73/F74 used in Paper X). (ii) Whether a clean paired construction exists for the massive sectors.

References

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4. P. Laurent *et al.*, “Constraints on Lorentz Invariance Violation using INTEGRAL/IBIS,” *Phys. Rev. D* **83**, 121301 (2011); *Fermi*/GBM and INTEGRAL gamma-ray polarimetry constraints on vacuum birefringence.
5. M. Ludwig, *Physics Notes* (2007), pp. 5–6 (“spinor electrodynamics; two $\gamma_{1/2}$ that only occur as a pair”).
6. Project findings: F69 (paired-spinor photon), F68 (minimal coupling forces the even photon), F67 (even-law vs bilinear mutually exclusive), F66/F65 (birefringence bounds; helicity–chirality map), F37 (RS/BCC chirality), F39 (two-helicity composite, retired as the photon), F91 (pairing classification).

Companion papers: I (substrate), IV (electromagnetism), VII (gravity), VIII (black hole / cell-size co-certification).